

Module 12: Darwin and Evolution — Keys to Success

Learning Objectives

By the end of this module, you should be able to:

1. Explain what Darwin meant by "descent with modification."
2. Describe the logical steps of natural selection (variation, overproduction, differential reproductive success).
3. List and explain the primary lines of evidence for evolution (fossils, biogeography, comparative anatomy, molecular biology).
4. Explain why *individuals do not evolve*, but *populations do*.

The Big Picture

"Nothing in biology makes sense except in the light of evolution." — Theodosius Dobzhansky

Evolution is the unifying thread of all biology. It explains both the immense diversity of life on Earth (how things are different) and the profound unity of life (why all life uses the exact same DNA code, as we saw in Module 07).

Key Terms & Concepts

1. Natural Selection (The "How" of Evolution)

Many people before Darwin thought species might change over time, but Charles Darwin (along with Alfred Russel Wallace) provided a logical mechanism for *how* it happens: **Natural Selection**.

It works like a simple logic puzzle based on three observations: - **Observation 1: Variation.** Individuals in a population vary in their traits (due to genetics). - **Observation 2: Overproduction.** More offspring are born than the environment can support, leading to competition. - **Conclusion: Differential Reproductive Success.** Individuals with traits best

suited to the current environment are more likely to survive and reproduce, passing those advantageous traits to the next generation.

Over many generations, the population adapts to its environment.

2. "Survival of the Fittest" – A Misunderstanding

The phrase "survival of the fittest" is often misunderstood to mean the biggest, strongest, or most aggressive survive.

In biology, **Evolutionary Fitness** has a very specific definition: *It is simply a measure of an individual's relative reproductive success.* - If a tiny, camouflage-colored insect hides well, survives, and has 100 offspring, it is *highly fit*. - If a massive, strong, brightly colored insect gets eaten by a bird before reproducing, its fitness is *zero*. - **Fitness = babies.**

3. Descent with Modification

Darwin didn't initially use the word "evolution." He called it "descent with modification." This phrase beautifully captures two ideas: 1. **Descent:** All living things share common ancestors (unity of life). 2. **Modification:** As descendants move into different habitats, they accumulate adaptations to those specific environments over millions of years (diversity of life).

4. Individuals Do Not Evolve!

This is a classic trap! Note the difference: - **Individuals acclimate or develop.** (If you lift weights, you get stronger muscles, but you won't pass those muscles to your children. Your DNA didn't change!) - **Populations evolve.** Evolution is the changing of genetic trait frequencies across an entire population over generations.

Evidence for Evolution

Scientists don't just "guess" evolution happens; it is supported by overwhelming evidence from multiple independent fields of science:

Line of Evidence	What it Shows	Example
Fossil Record	Shows the historical sequence of life and transitional forms.	Fossils showing the slow transition from land mammals to modern whales.
Comparative Anatomy	Homologous structures (similar bones in different animals) show modification from a common ancestor.	A human arm, a bat wing, and a whale flipper all have the exact same bone layout.
Biogeography	The geographic distribution of species aligns with geological history.	Marsupials are almost entirely isolated in Australia because the continent broke away early.
Molecular Biology	All life shares the same genetic code (DNA); closely related species have more similar DNA sequences.	Humans and chimpanzees share ~98.8% of their identical DNA code.

Study Tips

1. **Evolution is an editor, not a creator.** Natural selection can only act on trait variations that *already exist* in a population. It doesn't magically create what is needed on demand.
2. **Keep the environment in mind.** A trait that is highly advantageous in one environment might be a death sentence in another. Evolution is not "goal-oriented" toward a perfect being; it's simply adapting to the *current* local environment.